

AC9S3I01

AC9S3I02

AC9S3I03

AC9S3I04

AC9S3I05

AC9S3I06

Learning Intention: I can pose a testable question, make a prediction, and plan a fair test.**Success Criteria — I can:**

- Pose a question that can be investigated with a fair test.
- Make a prediction based on what I already know.
- Plan a fair test, changing only one thing at a time.

Strategy Box — What Makes a Test Fair?**Pose & Plan**

A **fair test** only changes **one thing** (the variable you're testing) while keeping everything else exactly the same. This way, you know that any difference in results is caused by the one thing you changed — not by something else.

Worked Example — Paper Plane Investigation

Stage	Example
Question (I01)	Does the shape of a paper plane affect how far it flies?
Prediction (I01)	"I think the pointed dart plane will fly furthest because it looks more streamlined."
What changes	The shape/design of the paper plane
What stays the same (I02)	Same paper size, same person throwing, same throwing strength, same location, no wind

Try It — Keep It Fair

A class is testing whether the colour of a paper plane affects how far it flies. Tick the things that should stay the **same** for this to be a fair test.

Item	Stay the same? (✓)
Colour of the paper	
Size and shape of the plane	
Person throwing the plane	
Throwing strength	

Watch Out!

If you change more than one thing at a time (like the shape and the paper size), you won't know which change caused the difference in results — the test is no longer fair.

Parent Tip

Fold two paper planes with your child using different designs, and test them together in a hallway or backyard. Talk about what needs to stay the same for it to be a fair comparison.

Key Vocabulary

Word	Meaning
Fair test	An investigation where only one thing is changed at a time
Variable	Something in an investigation that can change or be changed
Prediction	A sensible guess about the result, made before testing

Observe, Measure & Organise the Data

Q1 — Record the Measurements

Observe & Measure

The class measured how far (in metres) each plane design flew, across 3 trials, using a tape measure. Find the average for each design.

Design	Trial 1 (m)	Trial 2 (m)	Trial 3 (m)	Average (m)
Dart	7	9	8	
Glider	4	6	5	
Standard	6	5	7	

Tip: average = add the three trials together, then divide by 3.

Q2 — Choosing the Right Instrument

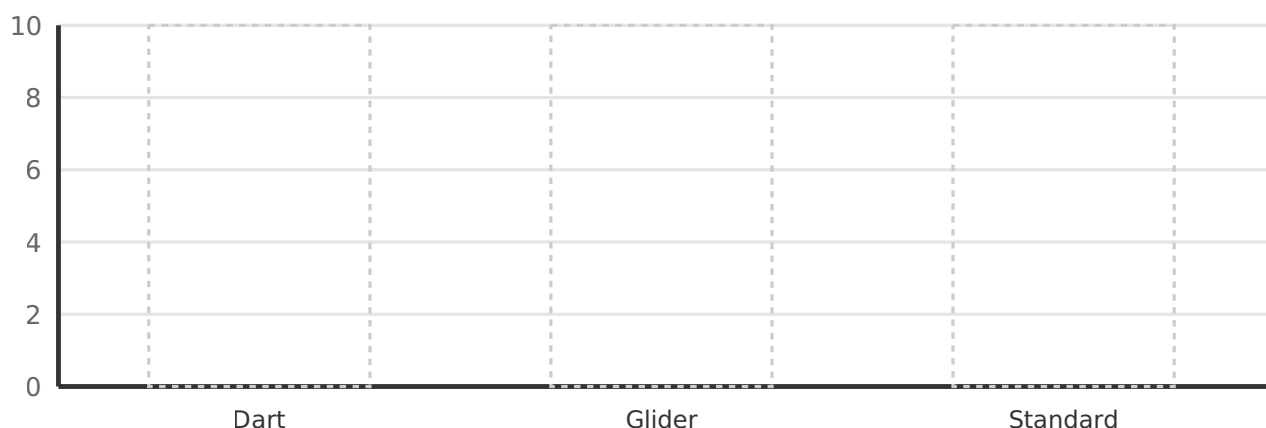
Circle the best measuring instrument for each task.

- Measuring how far a paper plane flies — ruler / tape measure / thermometer
- Measuring the temperature of water — tape measure / thermometer / stopwatch
- Measuring how long it takes a plane to land — ruler / stopwatch / scales

Q3 — Organise the Data into a Graph

Organise

Using your average distances from Q1, draw a bar for each plane design on the graph below.



Watch Out!

Always label your graph clearly — without labels, no one else can understand what your data shows, even if the measurements themselves are accurate.

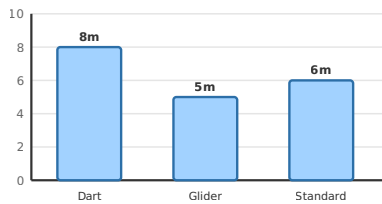
Reflection — Look at Your Graph

- Which plane design flew the shortest average distance? _____
- Which two designs are closest in average distance? _____

Compare, Conclude & Communicate

Worked Example — Results Graph

Compare



1 Compare: Was the class's prediction (that the Dart would fly furthest) correct? Use the graph to explain.

2 Conclude: Suggest one possible reason why the Dart flew furthest.

3 Communicate: Write a short findings report (2–3 sentences) explaining what the investigation found, for someone who wasn't there to see it.

Quick Check — Exit Ticket

Why is it important to compare your results to your original prediction?

Answer & Teacher Support

Answer Key — Pages 1 & 2

Try It (p1): Stay the same: size/shape of plane, person throwing, throwing strength. Changes: colour only.

Q1 (p2): Dart avg = $(7+9+8) \div 3 = 8\text{m}$. Glider avg = $(4+6+5) \div 3 = 5\text{m}$. Standard avg = $(6+5+7) \div 3 = 6\text{m}$.

Q2 (p2): a) Tape measure. b) Thermometer. c) Stopwatch.

Q3 (p2): Bars should show Dart=8m, Glider=5m, Standard=6m, matching Q1 averages.

Reflection (p2): a) Glider (5m). b) Glider and Standard are closest (5m vs 6m, only 1m apart).

Answer Key — Page 3

Q1: Yes — the Dart flew the furthest (8m average), matching the class's prediction.

Q2 (acceptable-answer guide): Accept any sensible reason, e.g. its pointed, streamlined shape may cut through the air with less resistance.

Q3 (acceptable-answer guide): Accept any accurate 2–3 sentence summary stating the Dart flew furthest on average, referencing the actual distances.

Exit ticket: Comparing to the prediction shows whether your idea was supported by evidence, which is how scientific understanding improves.

Teacher Checklist

- ☐ Poses a testable question and prediction (I01).
- ☐ Identifies what must stay the same in a fair test (I02).
- ☐ Records and averages measurements accurately (I03).
- ☐ Organises data into a correctly labelled graph (I04).
- ☐ Compares findings to the prediction and suggests a reason (I05).
- ☐ Communicates findings clearly for an audience (I06).

Teacher Tip

This single worksheet deliberately covers all six Science Inquiry codes (AC9S3I01–I06) as one connected investigation cycle, since these skills are naturally sequential. For maximum impact, run this as a real paper-plane experiment before completing the sheet.